

Welcome to Code Crunch!

C[6] | [CYBER CRUNCH: Ethical Hacking]

[Thu 1-2PM] Weekly | Spring 2025

Attendance



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What is Ethical Hacking?

- **Definition:** The practice of testing systems and networks for vulnerabilities with permission to improve security.
- **Key Phases:**
 - Information Gathering
 - Scanning
 - Exploitation
 - Privilege Escalation
 - Post-Exploitation
- **Types of Hackers:**
 - White Hat: Ethical hackers.
 - Black Hat: Malicious attackers.
 - Grey Hat: In-between.

Stages of Ethical Hacking

- **Information Gathering:** Collect publicly available information
 - Example: `whois`, `dig`, `nslookup` for domain information
 - `theHarvester` to gather emails, subdomains, and company data
- **Enumeration/Scanning:** Identify running services and applications
 - `nmap` for network scanning
 - `dirb` or `gobuster` for directory enumeration
- **Exploitation:** Take advantage of discovered vulnerabilities
 - `sqlmap` for SQL injection
 - `msfconsole` (Metasploit) for exploiting known vulnerabilities
- **Privilege Escalation:** Expand access to higher privileges
 - `linpeas.sh` to detect privilege escalation paths
 - `sudo -l` to check available sudo commands
- **Post-Exploitation:** Pivoting, data extraction, covering tracks
 - `netstat -ano` to identify further attack paths
 - `history -c` to clear shell command history

Industry-Standard Methodologies



- No single approach fits all situations
- Steps taken depend on the engagement scope
- General methodology stages
- Industry-standard methodologies (OSSTMM, OWASP, NIST, NCSC CAF)



wallarm



National Cyber
Security Centre

NIST



Importance of Ethical Hacking

- **Why It Matters:**
 - Prevents data breaches.
 - Identifies vulnerabilities before malicious actors do.
 - Builds trust with users and customers.
- **Real-World Example:** Target's 2013 data breach – Could have been mitigated with proactive ethical hacking.
- **Visual:** Timeline or flowchart showing the impact of ethical hacking.



Careers in Ethical Hacking

- **Popular Roles:**
 - Penetration Tester
 - Red Team Specialist
 - Bug Bounty Hunter
- **Certifications to Explore:**
 - Certified Ethical Hacker (CEH)
 - Offensive Security Certified Professional (OSCP)

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Hands-On Activity Overview

Platform: TryHackMe

- **Activity:** Ethical Hacking Fundamentals Room.
- **Concepts Covered:**
 1. Understanding attack surfaces.
 2. Performing reconnaissance and scanning.
- **Steps:**
 1. Follow instructor walkthrough for the first sections.
 2. Spend 15-20 minutes completing specific tasks independently.
 3. Share insights and challenges faced.
- **Visual:** Screenshot of the TryHackMe interface.

Interactive Activity: Penetration Testing Walkthrough



- **Objective:** Perform a mock penetration test on a simulated environment.
- **Setup:**
 1. Pre-configured virtual machine or browser-based tool.
 2. Access to a mock vulnerable server (e.g., DVWA or Metasploitable).
- **Steps:**
 1. Reconnaissance: Use tools like `ping` and `whois`.
 2. Scanning: Run `nmap` scans to identify open ports and services.
 3. Exploitation: Simulate a basic attack (e.g., brute-forcing a password).
- **Debrief:** Discuss findings and remediation strategies.
- **Visual:** Network diagram showing the simulated penetration test.

Debrief and Key Takeaways

- **Key Learnings:**
 - Ethical hacking is systematic and requires permission.
 - Real-world penetration testing helps prevent breaches.
 - Collaboration and communication are vital.
- **Next Steps:**
 - Explore additional ethical hacking tools.
 - Reflect on how ethical hacking applies to real-world careers.
- **Visual:** Checklist graphic or summary slide.



Q&A Session



Attendance



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Join us for the next sessions



Thursdays: 11AM – 12PM & 2PM – 3PM

Friday: 3PM – 4PM

Monday: 12PM – 1PM



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